

Lecture 12: ■ Definition of Computer Based Information System (CBIS)

A **Computer Based Information System (CBIS)** is a structured arrangement of people, hardware, software, data, and procedures that work together to collect, process, store, and distribute information for decision-making, coordination, control, analysis, and visualization in an organization.

🔑 **Key Components of CBIS**

- **Hardware:** Physical devices like computers, servers, and networking equipment.
- **Software:** Programs and applications that process and manage data.
- **Data:** Raw facts that are processed into meaningful information.
- **People:** Users such as managers, IT staff, and employees who interact with the system.
- **Procedures:** Rules and methods that govern how data is collected, processed, and used.

🎯 **Purpose of CBIS**

- To **support decision-making** by providing timely and accurate information.
- To **improve efficiency** in organizational operations.
- To **facilitate communication** and coordination among departments.
- To **store and secure information** for future use.

In short, a CBIS transforms raw data into useful information that helps organizations like the **University of Sindh, Jamshoro** manage academic, administrative, and research activities effectively.

Types of CBIS

- **Transaction Processing System (TPS)**
Handles day-to-day business transactions such as fee collection, student registration, payroll, and library management. It ensures accuracy and efficiency in routine operations.
 - **Management Information System (MIS)**
Provides summarized reports to managers for decision-making. For example, generating reports on student performance, departmental budgets, or attendance statistics.
 - **Decision Support System (DSS)**
Helps in making non-routine decisions by analyzing large datasets and providing simulations or models. For instance, predicting student enrollment trends or evaluating academic policies.
 - **Executive Support System (ESS)**
Designed for top-level management to monitor organizational performance and strategic planning. It provides dashboards and visual reports for quick insights.
 - **Office Automation System (OAS)**
Supports office activities like document management, scheduling, communication (emails, video conferencing), and collaboration tools.
 - **Knowledge Work System (KWS)**
Assists professionals and researchers in creating and sharing knowledge. Examples include research databases, academic publishing tools, and design software.
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Multiple Choice Questions (MCQs)

1. **CBIS Definition:**

A Computer Based Information System (CBIS) is primarily designed to:

- a) Play games
- b) Process and manage information
- c) Manufacture hardware
- d) Replace human intelligence

2. **CBIS Components:**

Which of the following is NOT a component of CBIS?

- a) Hardware
- b) Software
- c) People
- d) Weather

3. **Hardware:**

In CBIS, hardware refers to:

- a) Programs and applications
- b) Physical devices like computers
- c) Raw facts and figures
- d) Rules and procedures

4. **Software:**

Which component of CBIS converts data into meaningful information?

- a) Hardware
- b) Software
- c) People
- d) Procedures

5. **Data:**

Data in CBIS is best described as:

- a) Processed information
- b) Raw facts
- c) Rules and regulations
- d) Human input

6. **Procedures:**

Procedures in CBIS are:

- a) Physical devices
- b) Rules and methods
- c) Raw facts
- d) Software programs

7. **TPS:**

Which CBIS type handles day-to-day transactions like fee collection?

- a) MIS
- b) DSS

- c) TPS
 - d) ESS
8. **MIS:**
MIS provides:
- a) Raw data
 - b) Summarized reports for managers
 - c) Transaction details only
 - d) Entertainment services
9. **DSS:**
DSS is mainly used for:
- a) Routine transactions
 - b) Strategic planning
 - c) Non-routine decision making
 - d) Office automation
10. **ESS:**
ESS is designed for:
- a) Clerical staff
 - b) Middle managers
 - c) Top-level executives
 - d) Students
11. **OAS:**
Office Automation Systems (OAS) support:
- a) Document management and communication
 - b) Payroll processing
 - c) Strategic planning
 - d) Research databases
12. **KWS:**
Knowledge Work Systems are mainly used by:
- a) Clerks
 - b) Researchers and professionals
 - c) Executives
 - d) Students only
13. **CBIS Purpose:**
The main purpose of CBIS is to:
- a) Entertain users
 - b) Support decision-making
 - c) Replace managers
 - d) Eliminate data
14. **Efficiency:**
CBIS improves organizational efficiency by:
- a) Increasing manual work
 - b) Automating processes

- c) Reducing communication
- d) Ignoring procedures

15. **Communication:**

CBIS facilitates communication by:

- a) Blocking emails
- b) Providing tools like email and video conferencing
- c) Removing managers
- d) Limiting data storage

16. **TPS Example:**

Which of the following is an example of TPS in a university?

- a) Student registration system
- b) Research database
- c) Executive dashboard
- d) Email system

17. **MIS Example:**

A report showing student performance trends is generated by:

- a) TPS
- b) MIS
- c) DSS
- d) ESS

18. **DSS Example:**

Predicting student enrollment trends is a function of:

- a) TPS
- b) MIS
- c) DSS
- d) OAS

19. **ESS Example:**

A dashboard showing overall university performance is part of:

- a) TPS
- b) MIS
- c) ESS
- d) KWS

20. **OAS Example:**

Using Microsoft Word and Outlook in office work is an example of:

- a) TPS
 - b) OAS
 - c) DSS
 - d) ESS
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✓ **Answers with Explanation**

1. b) CBIS processes and manages information.
2. d) Weather is not a CBIS component.
3. b) Hardware = physical devices.
4. b) Software processes data.
5. b) Data = raw facts.
6. b) Procedures = rules/methods.
7. c) TPS handles routine transactions.
8. b) MIS provides summarized reports.
9. c) DSS supports non-routine decisions.
10. c) ESS is for executives.
11. a) OAS supports office communication.
12. b) KWS is for researchers/professionals.
13. b) CBIS supports decision-making.
14. b) CBIS automates processes.
15. b) CBIS provides communication tools.
16. a) Student registration = TPS.
17. b) Performance reports = MIS.
18. c) Enrollment prediction = DSS.
19. c) Executive dashboards = ESS.
20. b) Word/Outlook = OAS.